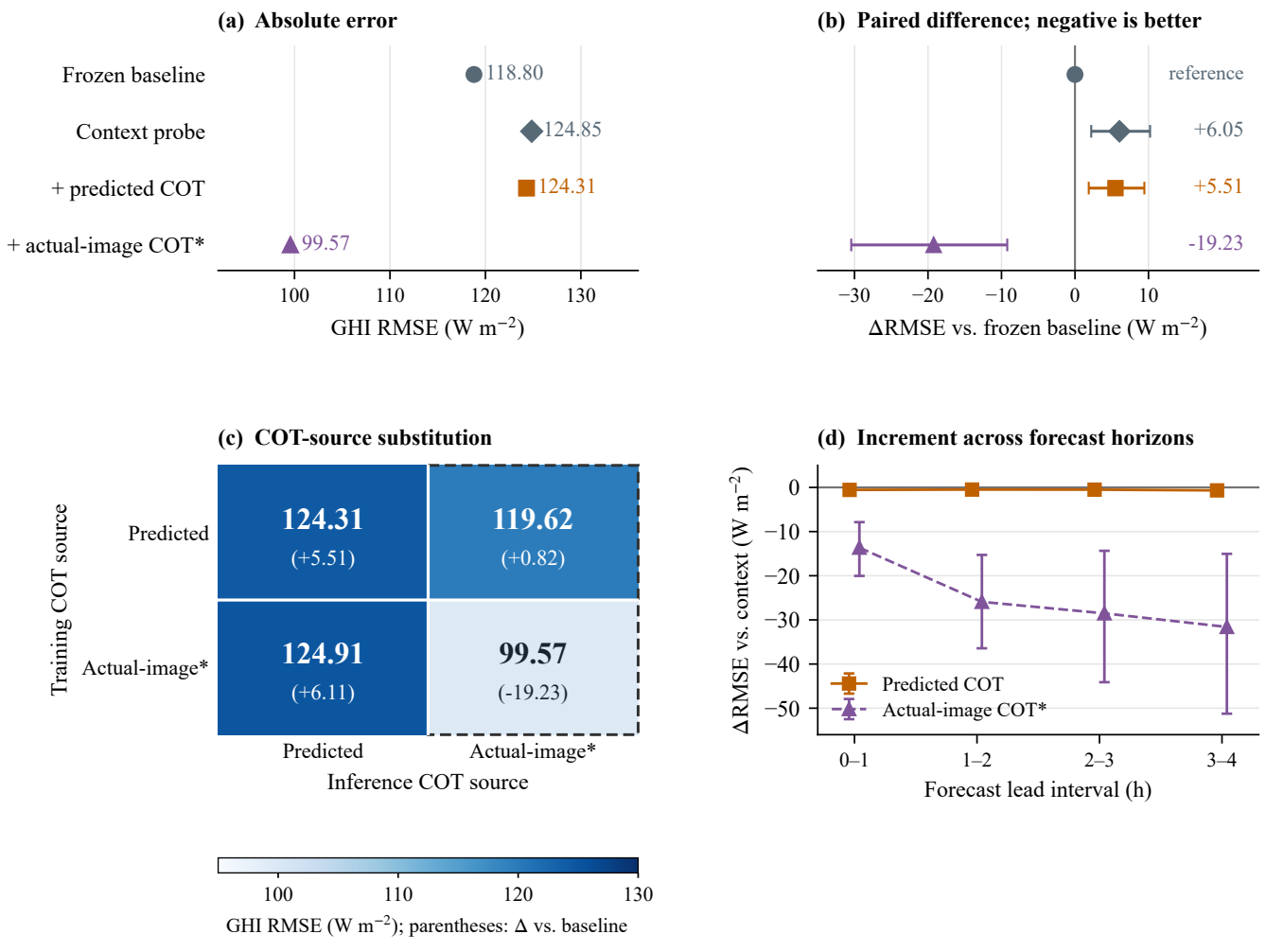

(a) Absolute error
Frozen baseline
Context probe
+ predicted COT
+ actual-image COT*
118.80
124.85
124.31
99.57
100
110
120
130
GHI RMSE (W m−2)
(b) Paired difference; negative is better
reference
+6.05
+5.51
-19.23
−30
−20
−10
0
10
ΔRMSE vs. frozen baseline (W m−2)
(c) COT-source substitution
Training COT source
Predicted
Actual-image*
124.31
(+5.51)
119.62
(+0.82)
124.91
(+6.11)
99.57
(-19.23)
Predicted
Actual-image*
Inference COT source
100
110
120
130
GHI RMSE (W m−2); parentheses: Δ vs. baseline
(d) Increment across forecast horizons
ΔRMSE vs. context (W m−2)
0
−10
−20
−30
−40
−50
Predicted COT
Actual-image COT*
0–1
1–2
2–3
3–4
Forecast lead interval (h)